



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Test 1: Review Topics

Test: Thursday 8 October, Period 1 — your last lesson before half term

Everything on the test is on this page. Tick a box when you can do it without looking at your notes.

1. Scalars and vectors

- ☐ State the difference: scalars have magnitude only (distance, speed); vectors have magnitude **and** direction (displacement, velocity, acceleration).
- ☐ Represent a vector as an arrow: length = magnitude, direction = direction.

2. Distance, displacement and speed

- ☐ State the difference between distance (how far, no direction) and displacement (straight-line distance from start to finish, **with** a direction).
- ☐ Recall typical speeds — walking ~ 1.5 m/s, running ~ 3 m/s, cycling ~ 6 m/s — and for transport: motorway ~ 30 m/s, fast train ~ 70 m/s, speed of sound ~ 330 m/s.
- ☐ Use $s = vt$ for constant speed, and calculate **average speed for non-uniform motion** (total distance $\div$ total time).

3. Velocity

- ☐ State that velocity is speed **in a given direction**, and explain the scalar–vector distinction for all four quantities: distance, displacement, speed, velocity.
- ☐ **(HT)** Explain why motion in a circle at constant speed still involves changing velocity, and is therefore accelerating.

4. Distance–time graphs

- ☐ Determine speed from the **gradient** of a distance–time graph.
- ☐ Draw a distance–time graph from measurements, and interpret its lines and slopes.

5. Velocity–time graphs and acceleration

- ☐ Use $a = \Delta v/t$, and state that an object slowing down is decelerating.
- ☐ Determine acceleration from the **gradient** of a velocity–time graph, and **(HT)** distance from the **area** underneath it — including finding that area by counting squares when the shape is not a simple rectangle or triangle.
- ☐ **Estimate the size of an everyday acceleration** — aeroplane on the runway ~ 3 m/s², car ~ 5 m/s², F1 car ~ 15 m/s².

6. The equation $v^2 - u^2 = 2as$

- ☐ Use and rearrange $v^2 - u^2 = 2as$, and know when to reach for it: velocity is changing, you are not given time, and you have no graph to find an area from.
- ☐ Do not mix up u (initial) and v (final) — v is not always the larger one.

Equations to learn by heart: $s = vt$ $a = \Delta v/t$

Given on the equation sheet: $v^2 - u^2 = 2as$

Practise with the Forces 1 Review Packet on your class page.